

Efrain Rocha — Senior Oracle DBA (Key Personnel)

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Technologies in use on this contract — where used

Technology	Direct use
Oracle database	Enterprise Oracle data warehouses integrated for trading, logistics, and operational reporting (JPMorgan Chase, 2010–14); Oracle Cloud Infrastructure partnership delivering a cloud laboratory platform (Gene by Gene) [FLAG: add versions administered and DBA duties from Efrain's reply]
Large-scale relational data platforms	PostgreSQL at 90+ TB, MySQL, MSSQL: partitioning, sharding, replication, retention strategies; led the database team (Gene by Gene, 2015–24)
AWS and Oracle OCI	Cloud platform delivery and operations; infrastructure-as-code; CI/CD pipelines
Java, Python, SQL, shell	Production engineering languages across 25 years; working C#/JavaScript
Messaging and service architectures	RabbitMQ request/response and asynchronous workflows; batch-to-message migrations; legacy API replacement
Docker and CI/CD	Platform reliability engineering and pipeline ownership in regulated environments [FLAG: confirm specifics]

Proposed role

Senior Oracle DBA: owns the CWMS schema changes the authorization design requires (embargo table, mask-view extension, `cwms_sec` procedure changes), reviews the interaction between CDA-level filtering and Oracle Virtual Private Database policies, keeps the integration-test database images current with schema pull requests, and advises on Oracle RDS behaviour in AWS GovCloud including the write-burst throttling the Government has observed.

Experience

U.S. LawShield — Senior Development Manager | Platform Architect · Houston, TX · 2024 – present

- Modernized a revenue-impacting backend by replacing a decade-old tightly coupled API layer with a service-oriented, message-driven architecture; improved response times, reliability, and maintainability; decoupled core systems with RabbitMQ request/response messaging so services scale and fail independently.

Gene by Gene — Senior Development Manager | Enterprise Architecture · Houston, TX · 2015 – 2024

- Built and led the engineering, DevOps, and database teams for high-throughput genomic and laboratory platforms under strict regulatory and delivery constraints.

- Reduced DNA analysis turnaround from ~30 days to ~5 days and core processing from ~6 hours to ~20 minutes; delivered a distributed processing platform backed by a 90+ TB PostgreSQL data platform.
- Partnered with Oracle OCI to deliver a cloud-based laboratory platform for time-critical, high-volume diagnostic processing.
- Re-engineered batch and file-based pipelines into resilient message-driven workflows; operated 100+ VMs continuously to re-analyze 1.6 million DNA samples.
- Led a critical three-week delivery adapting analysis systems after a lab technology change — work normally taking two to three months.

Audubon Engineering — Senior Manager, CAD/CAE Software Applications · Houston, TX · 2014 – 2015

- Delivered and stabilized internal engineering platforms across upstream, midstream, downstream, and pipeline operations; directed application architecture and integrations.

JPMorgan Chase & Co. — Vice President, Physical Logistics Core · Houston, TX · 2010 – 2014

- Led delivery of a reusable physical-logistics platform supporting natural-gas trading across the Americas and EMEA; directed the architecture of a shared logistics framework; integrated with Endur and enterprise Oracle data warehouses for trading, logistics, and operational reporting.

Earlier leadership roles · 2001 – 2007

- Director, Product & Architecture — Codefast, Inc.; Director of Engineering & Quality Assurance — SunGard Energy; Director, Enterprise Systems Architecture — El Paso Global Networks. Led Director-level engineering organizations delivering long-lived enterprise platforms in telecom and energy, including systems at WorldCom/MCI and Verizon that remained operational for more than twenty years.

Education

M.S., Computer Science — Monterrey Institute of Technology (ITESM). B.S., Computer Science Engineering — Monterrey Institute of Technology (ITESM).

Certifications and special qualifications

[FLAG: Oracle certifications, if any, from Efrain's reply — otherwise state "none; qualification rests on 25+ years of production database engineering"]. U.S. citizen.